

Magnetic Tunnel Junctions for Timekeeping in Intermittent Computing Systems

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Abstract

Batteryless intermittent systems run unattended for years, but power failures erase timekeeping state, corrupting sensing, scheduling, and coordination. State-of-the-art timekeepers infer elapsed time from capacitor discharge; however, the capacitor must be sized for the longest interval measured (so range, energy, and area grow together), and repeated charge–discharge cycling lowers capacitance over time, biasing every estimate further as the deployment ages. We present FLINT, a timekeeper that reads elapsed time from the stochastic retention loss of an array of “broken” Magnetic Tunnel Junctions (MTJs)—spintronic memory cells engineered to lose state predictably. Because the decay timescale is fixed by device geometry, the energy to read it is independent of the interval measured and does not drift with device age. We validate FLINT’s array model against 21 fabricated MTJs, then evaluate the full timekeeper in real-device-trace-driven simulation, showing that it tracks **over 15 minutes** of off-time within 10% error while consuming **only 1.03 μ J** and occupying **under 0.1 mm²**—**9.2 $\times$** the range at **11 $\times$** lower energy than prior work. It extends to longer intervals at no added cost, and makes 16–52 $\times$ fewer scheduling errors than an aging capacitor clock over a one-year deployment.

1 Introduction

Liberating embedded sensing devices from batteries and powering them entirely with energy harvested from ambient sources—solar [34], RF [56], and thermal [9]—has unlocked applications previously out of reach [15, 17, 18, 22]. Harvested energy accumulates in a small capacitor; the device runs only while this energy is available and powers down once it is depleted [24, 66], repeatedly cycling between execution and complete power loss. This makes frequent power failures the norm for these devices, and so computing becomes *intermittent*. To ensure forward progress, intermittent computing runtimes preserve computation state [2, 10, 59], memory consistency [52, 71], and peripheral context [12, 67] across power failures, allowing execution to resume once energy returns.

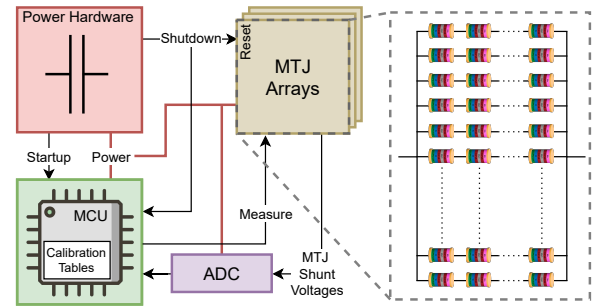


Figure 1: FLINT timekeeper system overview and dataflow. Instead of capacitive decay, FLINT relies on broken memory (MTJ arrays) tuned to lose data predictably during power failures. On power return, a simple resistance-based array readout measures elapsed time.

With no batteries to service, these systems can be used in long-term *unattended* deployments. However, while existing runtimes preserve system state, they cannot preserve temporal continuity, yet many intermittent applications require time-sensitive operations such as periodic sensing, timeout management, and event scheduling [29, 37, 38, 42, 46, 53, 67, 68]. For example, wildlife tracking collars periodically record GPS locations every few seconds [44], cold-chain monitoring systems sample temperature every few minutes to detect spoilage [57], and infrastructure monitoring platforms schedule strain measurements days apart [1]. Consequently, the required timekeeping interval varies by orders of magnitude across deployments, ranging from seconds to minutes to days. In all cases, losing track of elapsed time can result in missed sensing events, premature actions, stale data, or incorrect timeout decisions that can compromise system correctness for the lifetime of an unattended deployment. Unpredictable ambient energy availability further exacerbates this problem, as it causes power interruptions to vary significantly in timing and duration.

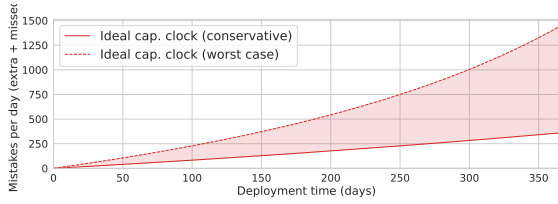


Figure 2: Daily capacitor clock mistakes over 1 yr. Bands: conservative (20 %/yr) to worst-case (50 %/yr) degradation. See Sec. 5.4, Sec. 6.6.

Together, these factors make reliable timekeeping a fundamental requirement for batteryless systems. We therefore identify three requirements for a practical batteryless timekeeper:

- R1 Application-agnostic operation.** *A timekeeping solution must be generalizable, not bespoke*—independent of application-specific assumptions, workloads, energy sources, or deployment environments. Its energy and latency overhead must remain flat across all operating ranges—short intervals to multi-hour gaps alike—without hardware modifications or re-sizing.
- R2 Temporal stability.** *A timekeeper must stay accurate across the full operational lifetime* of the device, not just at deployment. Component aging causes slow, cumulative drift—despite rated accuracy at deployment, the inaccuracies will only increase over time—and correcting it requires recalibration, which is impractical in most remote deployments.
- R3 Environmental robustness.** *A timekeeper must remain accurate despite fast-changing conditions* (e.g., supply voltage variation, temperature swings) that shift its readings within a single power cycle. Unlike aging, these perturbations are immediate and reversible, but no less damaging if uncompensated.

Existing capacitor-based timekeeping approaches [7, 20, 23, 39] fail to meet all three requirements. First, they assume capacitor discharge is a stable process. Charge–discharge cycling can degrade capacitance by up to half within a year [16], which shifts the discharge curve and consistently increases the error in every future estimate (R2). Fig. 2 shows missed and redundant samples accumulating as the capacitor ages. This drift is directional and cannot be corrected without recalibration, making capacitor clocks a futile attempt for unattended multi-year deployments. Furthermore, range and cost are correlated: a capacitor must be sized for the longest measurement interval and recharged each cycle; range, energy, area, and startup latency grow in lockstep [20, 23, 36]. Reaching a range of 15 minutes requires $\approx 101 \mu\text{J}$ of energy (Fig. 8)—on the same order of magnitude as some main system capacitors [35].

Second, capacitor-based timekeepers must be calibrated to specific workloads, harvesting conditions, and discharge profiles, limiting generality (R1). Larger off-time ranges demand larger capacitors, which charge more slowly, thereby increasing startup latency and reducing the fraction of events the device can capture. Lastly, capacitor discharge shifts with supply voltage and temperature, requiring recalibration after any environmental change (R3); correcting aging-induced drift demands recalibration too (R2). Both are infeasible for devices deployed for years in hard-to-reach areas.

Contributions: We take a fundamentally different approach: measuring time from device physics rather than stored energy. We leverage MTJs, spintronic memory cells engineered to “fail” at retaining data in a controlled, stochastic manner. Unlike conventional non-volatile memory, where retention loss is a defect, these predictable failures produce a stable statistical decay across an MTJ array. Reading this decay reveals elapsed time directly from device behavior, decoupling timekeeping cost from the interval measured.

We introduce Faulty Logic for Intermittent Nanojoule Timekeeping (FLINT), comprising MTJ devices, supporting circuitry, and a lightweight runtime kernel. FLINT (Fig. 1) organizes MTJs into arrays that generate a stable decay signal (R2) mapped to elapsed time. Time estimation uses simple ADC measurements with lightweight calibration and temperature correction (R3). FLINT tracks 938 s (over 15 minutes) of off-time within 10% geomean error while consuming $1.03 \mu\text{J}$. It occupies less than 0.1 mm^2 and adds $50 \mu\text{s}$ of latency to reset arrays at power-off and $70.83 \mu\text{s}$ to estimate time at power-on. This range (938 s) is not a hard limit: an array’s range is set by its device geometry, not stored energy, so FLINT extends to longer intervals without spending more energy or area—something a capacitor cannot do. Arrays draw no power when idle and occupy little area, so a single chip can hold many and select the appropriate range at runtime (R1). FLINT makes the following contributions:

- **Aging as an overlooked failure mode.** We identify capacitor aging as a critical but largely ignored source of timekeeping error in long-lived batteryless deployments, and quantify its effect: an aging capacitor clock accumulates 16–52 $\times$ more scheduling errors than FLINT over a one-year deployment.
- **A physics-based timekeeping primitive.** We show that broken MTJs reveal elapsed time through stochastic retention loss, giving the first timekeeper whose energy cost is independent of the measured duration and whose accuracy is immune to charge-cycle aging by construction.
- **The FLINT system.** We realize this primitive with multiple MTJ arrays, simple control hardware, lightweight calibration, temperature correction, and estimate fusion. Idle arrays consume no energy and occupy minimal area, so one chip can host more arrays than needed and let applications select their range at runtime—a “one-size-fits-all,” near-instant startup timekeeper.
- **Hardware-grounded evaluation.** We validate our array model against 21 fabricated MTJs and, in simulation (built and validated using fabricated MTJs and modeled circuits), show FLINT tracks over 15 minutes within 10% error at $1.03 \mu\text{J}$, holds its range as it ages, and remains accurate across temperatures. Five application case studies span two families: three accuracy studies (GPS tracking, structural health monitoring, cold-chain logistics) that quantify scheduling mistakes under capacitor aging, and two long-duration scenarios that capacitor clocks cannot cover at all.

2 Background and Motivation

Batteryless energy-harvesting devices operate on intermittent power cycles, utilizing harvested energy to drive computation whenever it becomes available. As shown in Fig. 3, energy accumulates in a storage capacitor. The device turns on once the voltage exceeds a predefined threshold (V_{on}), executes until the stored energy is depleted, and turns off when the voltage drops below the minimum

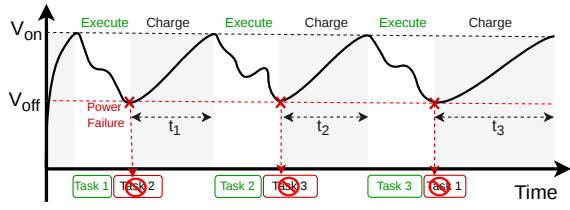


Figure 3: Intermittent system power curve. Note off time variability.

operating threshold (V_{off}). Due to variability in environmental energy sources, both charging time and execution duration are highly irregular, ranging from milliseconds to several minutes. As a result, application execution is repeatedly interrupted across multiple power cycles. This intermittent behavior introduces fundamental challenges for time-sensitive operations. Data collected in one execution phase may become outdated by the time it is processed or transmitted in a later cycle, as the elapsed real time between these phases is unknown. Without a notion of time across power failures, the system cannot determine data freshness or enforce temporal constraints. Consequently, persistent timekeeping becomes essential to support correct and energy-efficient execution of time-dependent tasks.

2.1 Persistent Timekeeping

Existing methods for persistent timekeeping can be categorized into three main types: RTCs, software-based, and capacitor-based methods. RTCs provide accurate timekeeping but require a continuous power source, limiting their applicability in fully batteryless settings. Software-based approaches infer time from system behavior, such as SRAM data remanence [40], oscillator stabilization [3], or ambient power levels [73], but are often sensitive to environmental conditions [73] and offer limited measurement ranges [3, 40].

Consequently, many systems utilize capacitor-based timekeeping [7, 20, 23, 70], which estimates elapsed time during power failures by monitoring the discharge of dedicated capacitors. In this approach, a capacitor is charged before a power failure and then allowed to discharge while the system is off, as shown in Fig. 4. When power is restored, the voltage level in the timekeeping capacitor is measured via an ADC on an MCU to calculate the elapsed time using known discharge time characteristics, as shown in Eq. 1.

$$t_{dchrg} = RC \ln(V_{init}/V_{discharge}) \quad (1)$$

V_{init} represents the initial voltage before power failure, and $V_{discharge}$ is the voltage across the timekeeping capacitor when the system reboots. The measurable time range is highly dependent on the capacitor size; larger capacitors can measure longer durations but require more energy, while smaller capacitors consume less energy but reduce the measurable time range. Additionally, unpredictable power conditions add noise and reduce accuracy.

Furthermore, larger capacitors increase the start-up time following each power failure, as the system must recharge the timekeeping capacitor to prepare for the next outage. Yet larger capacitors are

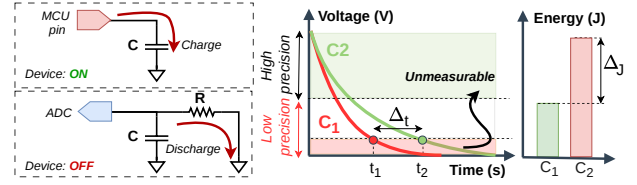


Figure 4: Capacitor-based timekeeping charges a capacitor during the on-time and discharges it during the off-time, with measurable duration and energy consumption determined by capacitor size.

exactly what low-power conditions demand to measure longer durations. Multiple-capacitor schemes [7, 20, 23] add flexibility by combining capacitors of different sizes, but they do not escape the trade-off: every additional second of range is paid for in energy, board area, and startup latency, and costs grow with target range.

A second limitation is that the capacitor degrades with use. Repeated charge–discharge cycling reduces capacitance by as much as 50% in a year [16]. A smaller C shortens the RC time constant (Eq. 1), so the capacitor discharges faster; the same elapsed time yields a lower measured voltage, which the calibrated mapping reads as a longer duration—the clock overestimates progressively as it ages. This error has a fixed sign, so unlike measurement noise it does not average out but accumulates over power cycles, skewing the time estimate over the deployment lifetime (Fig. 2). Recalibration could correct it, but unattended deployments offer no such opportunity.

Both limitations are structural: measuring time with stored energy ties range to both energy and area, and ties reliability to a component that wears out. This motivates a timekeeping primitive built on a stable physical process, whose cost is independent of the measured interval and whose behavior does not drift with age.

2.2 A Primer on MTJs

A magnetic tunnel junction (MTJ) is a nanoscale spintronic device comprising two ferromagnetic layers separated by a thin insulating tunnel barrier (see Fig. 5a). The reference layer maintains a fixed magnetic orientation, relative to which the free layer occupies one of two stable magnetic alignment states: parallel (P) or antiparallel (AP). Due to quantum mechanical tunneling across the barrier, the junction’s resistance depends strongly on this alignment—the P state exhibits lower resistance (R_P) and the AP state exhibits higher resistance (R_{AP}). This phenomenon is represented by the tunneling magnetoresistance ratio, $TMR = (R_{AP} - R_P)/R_P$, which for the devices characterized in this work averages 125% (see Fig. 5b).

The thermal stability of the free layer is governed by an energy barrier E_b , which determines the timescale for spontaneous state transitions. Switching behavior follows the Néel-Arrhenius relation, which states the mean dwell time (τ) in a given state scales as

$$\tau = \tau_0 \cdot \exp(E_b/k_b T) \quad (2)$$

where τ_0 is the attempt frequency, k_b is the Boltzmann constant, and T is temperature in Kelvin. By controlling E_b during fabrication, MTJs can be engineered with dwell times spanning many orders of magnitude. Devices with sufficiently low energy barriers switch stochastically under ambient thermal fluctuations alone, without

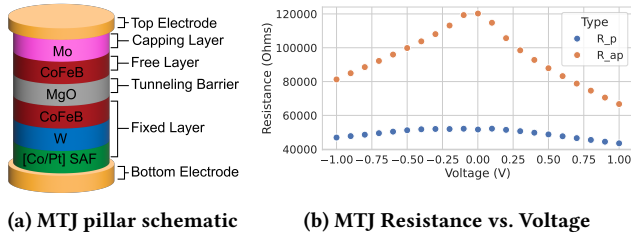


Figure 5: a) MTJ device stack. b) Resistance vs. applied voltage V for P (blue) and AP (orange) states. Note drop in TMR with increasing V .

requiring any applied write current [50]. Crucially, τ is fixed by the device geometry and material stack, not by any stored charge—so the timescale neither costs energy to maintain nor drifts as the device is cycled, directly addressing the two failure modes above.

While individual MTJ switching events are inherently random, large arrays of identically configured MTJs exhibit well-characterized aggregate behavior. The energy barriers across devices in an array follow a near-normal distribution (Fig. 11a), and as devices stochastically transition between states, the aggregate array imbalance ($|On - Off|$) and resistance decay in a smooth, reproducible, approximately exponential profile (Fig. 7a and Fig. 7b, respectively). By leveraging this statistical regularity, FLINT infers the elapsed time from a single resistance measurement of the array.

3 System Design

FLINT (Fig. 1) is a batteryless timekeeping system that estimates off-time by leveraging the stochastic decay of MTJ arrays, whose resistance changes as devices transition between P and AP states. Each array is reset to a known state before power loss and decays passively during outages, requiring no active energy.

3.1 FLINT Architecture and Data Flow

Fig. 6 illustrates the architecture of FLINT. The system consists of multiple MTJ arrays, an ADC for resistance measurement, and a software layer running on the MCU. Each MTJ array is accessed through simple control circuitry that enables reset and resistance measurement. As shown in Fig. 1, a shutdown signal resets the array before power loss, while a measurement signal enables ADC-based resistance sampling upon power restoration. The measured resistance is mapped to elapsed time using a lookup table stored in memory, constructed during calibration. To improve robustness and extend the timekeeping range, FLINT employs multiple arrays with different dwell-time characteristics. Their outputs are combined using lightweight fusion logic to produce the final time estimate. Additional correction modules, such as temperature compensation, can be applied at runtime with minimal overhead, maintaining accuracy under varying environmental conditions (R3).

3.2 Timekeeping Operation

FLINT estimates elapsed time across power failures through a sequence of operations spanning shutdown, off-time, and startup. Prior to power failure, each MTJ array is reset to a known initial

state, where all devices are in the same magnetic state. During the off-time, the arrays evolve passively without energy consumption. Individual MTJs stochastically switch between P and AP states due to thermal effects, causing the imbalance between states to gradually decrease over time. As shown in Fig. 7, this reduction in imbalance (Fig. 7a) drives a corresponding change in the overall array resistance (Fig. 7b), which evolves monotonically as the system approaches equilibrium. Different arrays exhibit different decay rates depending on their dwell-time characteristics, enabling the system to capture a wide range of time scales.

Upon power restoration, the system measures the resistance of each array using an ADC. This measured resistance is then mapped to elapsed time using a precomputed lookup table created in calibration, exploiting the monotonic relationship between resistance and time. Finally, when multiple arrays are used, their individual time estimates are combined using fusion logic to produce a single, robust estimate that balances resolution and timekeeping range.

3.3 Design Choices

Multi-array design. A single MTJ array provides limited timekeeping range due to saturation as devices approach equilibrium. To address this, FLINT employs multiple arrays with different dwell-time characteristics, enabling each array to capture a distinct portion of the time spectrum. Short dwell-time arrays provide fine resolution at small durations, while long dwell-time arrays extend the measurable range. This design allows FLINT to simultaneously achieve high resolution and long timekeeping range, which is difficult to realize with a single array.

Fusion across arrays. Combining estimates from multiple arrays is non-trivial due to differing stochastic dynamics and time sensitivities. Naïve averaging assumes a uniform and stationary behavior across arrays, which leads to large estimation errors. Instead, FLINT employs fusion mechanisms that prioritize arrays operating in their sensitive regions while discounting those that have saturated. This allows the system to dynamically balance short-term resolution and long-term accuracy, resulting in a more robust time estimate. **Lookup-based time inference.** FLINT maps measured resistance to elapsed time using a precomputed lookup table derived during calibration. While analytical models could be used, the stochastic nature of MTJ switching and device variability makes closed-form modeling challenging and less robust. A lookup-based approach naturally captures these effects and enables efficient runtime estimation with minimal computational overhead.

Passive timekeeping. FLINT performs timekeeping entirely through passive device dynamics during off-time, without requiring active energy consumption. Unlike capacitor-based approaches that rely on charging dynamics and active measurement circuitry, FLINT leverages intrinsic stochastic switching behavior to encode time directly in device state. This enables zero energy overhead during outages and eliminates the need for dedicated timing hardware.

Range decoupled from energy. A capacitor's range is set by how much charge it stores, so a longer range means a larger capacitor, more energy to charge it every cycle, and more board area—with energy scaling proportional to range (Fig. 8). FLINT breaks this coupling. Because elapsed time is derived from decay fixed by E_b , not stored charge, the energy to reset and measure an array is the

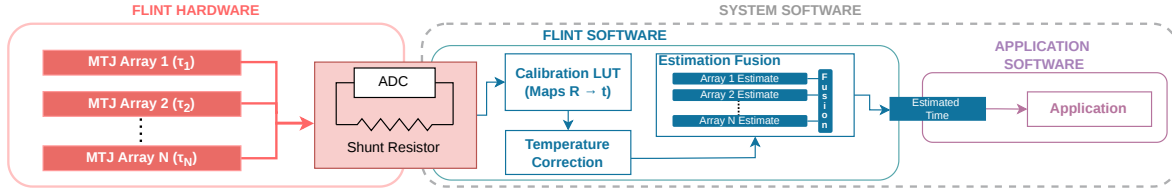
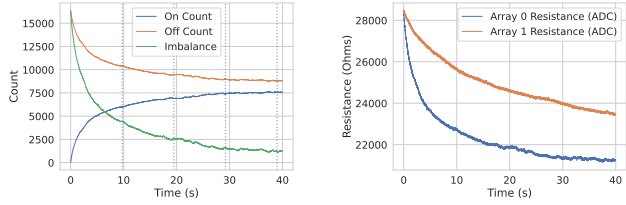


Figure 6: System architecture and dataflow. After the fusion stage, we obtain a single time estimate that is our result.



(a) 9.7 s dwell time array. Green line is $\text{imbalance} = |\text{On} - \text{Off}|$. Vertical dashed lines at dwell time multiples. (b) 9.7-second (blue) and 72-second (orange) dwell time arrays. Starting resistance is 28 k Ω .

Figure 7: MTJ array decay curves (64×256 arrays).

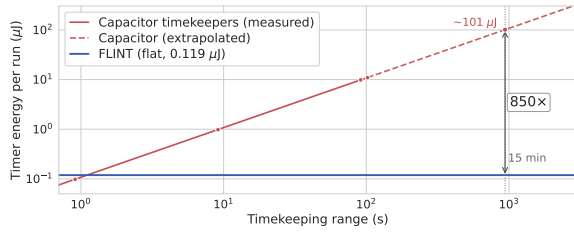


Figure 8: Timer energy vs. timekeeping range (log-log).

same whether it tracks seconds or hours. Reaching a 15-minute range costs a capacitor clock roughly 101 μJ of timer energy and a bulky discrete part; FLINT pays 0.119 μJ in any range (850 $\times$ less, see Sec. 6.4). This is also what makes FLINT a “one size fits all” timekeeper: since an idle array draws no power and a 64×256 array occupies just 655.36 μm^2 , a chip can carry more arrays than any deployment needs, and the application selects which to use at runtime—trading range for resolution without changing hardware.

Together, these properties directly address **R1** (application-agnostic operation): FLINT’s energy and footprint are independent of the target interval—short gaps or multi-hour durations—so the same hardware serves any workload without re-sizing or re-configuration. Aging immunity satisfies **R2** (temporal stability): because E_b is a device-geometry parameter unaffected by charge cycling, the decay timescale remains stable over the full deployment lifetime—no recalibration is needed as the device ages.

4 Implementation

FLINT consists of MTJs connected in a series-parallel configuration, surrounded by support hardware (Fig. 9). Time is inferred from the measured resistance through a software pipeline, shown in Fig. 6.

4.1 Control Hardware

We choose spin-transfer torque (STT) MTJs for FLINT because they are current-controlled and can be deterministically reset into a desired state, enabling reliable initialization. In addition, their τ can be controlled with adjustment of E_b . STT devices are CMOS-compatible and energy-efficient, making them practical for scalable array fabrication and implementation [65]. The control hardware for a single MTJ array is depicted in Fig. 9. It consists of the reset and measurement circuitry. The former **resets** and initializes the array just before shutdown, and is triggered when the system power reaches a predefined lower threshold. The latter is used to **measure** resistance of the array at startup, and is controlled by the SoC. These circuits are designed to minimize the required support hardware.

Reset Circuitry: A current pulse exceeding a pre-defined threshold (e.g. 35 μA for 50 ns) is used to reset the MTJ array to the AP state. To generate this pulse, we use an appropriately sized NMOS transistor for each row of the array (shown in blue in Fig. 9). The transistors are biased to operate in the saturation region and function as current sources, ensuring a consistent supply to each row. Since the MTJ resistance depends on the state and can differ across rows, each row requires a transistor to drive it.

Measurement Circuitry: To measure the array resistance, we determine the voltage across a series *shunt resistor* (R_{shunt}), use it to compute the current through the shunt and MTJ array, and thus infer the array resistance. The shunt resistance is chosen to be approximately equal to the array resistance to maximize ADC dynamic range. One ADC channel is required per array. The system VDD is significantly lower than the reset voltage, limiting the measurement current. An NMOS transistor controls the circuit and is sized to minimize current through the array, avoiding unintended MTJ switching or degradation of TMR (see Fig. 5b).

4.2 Calibration

Calibration is a vital process for ensuring high accuracy in varying deployment environments. To calibrate an array, it is first reset to its initial state. As the array decays, the resistance is sampled at a fixed interval, chosen as a small fraction of the dwell time (e.g., ≈ 0.05 –0.1%), to accurately capture the stochastic switching behavior of fast devices. These measured values of resistance are associated with their respective elapsed time, and the process is repeated. Finally, the resistance measured across these runs is averaged at

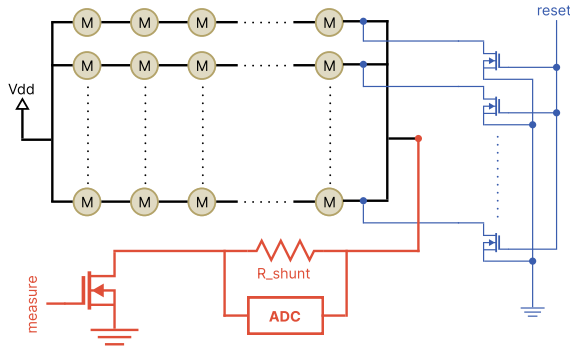


Figure 9: Supporting circuitry for a single array. Reset (blue), measurement (orange). Width = series, height = parallel.

each timestep. This is used to build the lookup table that holds the information needed to infer time. Note the sampling process can be done in parallel for all arrays with a sufficiently fast ADC.

Algorithm 1 Array Calibration Procedure

Require: Sampled data $\mathcal{D} = \{(t_1, R_1), \dots, (t_N, R_N)\}$, Table Size M

- 1: Sort $\mathcal{D}$ by time t , set $\mathbf{t} \leftarrow [t_1, \dots, t_N]$ and $\mathbf{R}_{\text{raw}} \leftarrow [R_1, \dots, R_N]$
- 2: $\mathbf{R}_{\text{iso}} \leftarrow \text{IsotonicRegression}(\mathbf{t}, \mathbf{R}_{\text{raw}})$ $\triangleright$ Filter noise, enforce monotonicity
- 3: $P \leftarrow \text{PCHIP}(\mathbf{t}, \mathbf{R}_{\text{iso}})$ $\triangleright$ Construct continuous mapping
- 4: $\mathbf{R}_{\text{LUT}} \leftarrow \text{LinSpace}(\min(P), \max(P), M)$
- 5: **for** $m = 1$ **to** M **do** $\triangleright$ Invert PCHIP output to build calibration table
- 6: $t_{\text{LUT}}[m] \leftarrow \text{Solve}(P(t) = \mathbf{R}_{\text{LUT}}[m])$
- 7: **return** Calibration LUT ($\mathbf{R}_{\text{LUT}}, t_{\text{LUT}}$)

Creating the lookup table (LUT) requires inverting the time vs. resistance data. We show our array calibration procedure in [Alg. 1](#). Isotonic regression (L2) [14, 21, 47] is used to fit a monotonic step function on the collected data, enabling inversion. PCHIP interpolation [31, 58] is applied to smooth the curve (L3), chosen as it preserves monotonicity. Finally, the LUT is built by creating a linear space of resistance values and sampling the PCHIP curve with them (L4-6).

The closest resistance value in the lookup table is found using a binary search. Linear interpolation is then used to compute the resulting time estimate. If the resistance value falls outside the table, it is clamped to the closest boundary.

4.3 Temperature Correction

To correct for temperature variations, we use the Néel-Arrhenius equation to scale our estimates based on the operating temperature T_{op} . Given the design energy barrier value $\hat{E}_b$ (as a temperature-dependent dimensionless quantity, rather than units of kT) and calibration temperature T_{cal} , the corrected time estimate t_{corr} is computed from the raw time estimate t_{raw} as follows:

$$t_{\text{corr}} = t_{\text{raw}} \cdot \exp\left(\hat{E}_b \cdot \left(\frac{1}{T_{\text{op}}} - \frac{1}{T_{\text{cal}}}\right)\right) \quad (3)$$

4.4 Estimate Fusion

Combining time estimates from each array must account for the variation in range and accuracy inherent to each array. A simple

Algorithm 2 Shortest Active Fusion

Require: Estimates $T = [t_1, \dots, t_N]$, Dwell times $D = [\tau_1, \dots, \tau_N]$

- 1: $G \leftarrow 1.25$ $\triangleright$ Reliability gate multiplier
- 2: $\tau_{\text{min}} \leftarrow \infty$, $idx_{\text{best}} \leftarrow \arg \max_i D[i]$ $\triangleright$ Fallback to longest dwell
- 3: **for** $i = 1$ **to** N **do**
- 4: $\text{update } idx_{\text{best}} \leftarrow i$, $\tau_{\text{min}} \leftarrow \tau_i$ **if** $t_i < G \cdot \tau_i$ **and** $\tau_i < \tau_{\text{min}}$
- 5: **return** $t_{idx_{\text{best}}}$

average of the estimated time does not capture these differences. To resolve this, we implement two fusion algorithms. The first chooses the shortest dwell time array that is still valid (i.e., the imbalance is still high), as shown in [Alg. 2](#). This is based on the fact that arrays with shorter dwell times can measure at a higher resolution and maximize accuracy. To determine if an array is valid, we compare its estimate t_{est} to its dwell time τ : if $t_{\text{est}} < G \cdot \tau$, the array is considered valid. The gate factor $G = 1.25$ identifies the optimal range for each array. If none of the arrays are deemed valid, the estimate from the array with the longest dwell time is used.

Algorithm 3 Linear-Decay Fusion (Linear & Sqrt)

Require: Estimates $T = [t_1, \dots, t_N]$, Decay cutoff factor $C = 2$, Dwell times $D = [\tau_1, \dots, \tau_N]$, Mode $M \in \{\text{Linear}, \text{Sqrt}\}$

- 1: $W_{\text{total}} \leftarrow 0$, $S_w \leftarrow 0$, $idx_{\text{longest}} \leftarrow \arg \max_i D[i]$
- 2: **for** $i = 1$ **to** N **do**
- 3: $s_i \leftarrow \tau_i$ **if** $M = \text{Linear}$ **else** $\sqrt{\tau_i}$ $\triangleright$ Scale factor
- 4: $gate_i \leftarrow \max(0, 1 - t_i/(C \cdot s_i))$, $w_i \leftarrow gate_i/s_i$
- 5: $W_{\text{total}} \leftarrow W_{\text{total}} + w_i$, $S_w \leftarrow S_w + w_i \cdot t_i$
- 6: **return** $t_{idx_{\text{longest}}}$ **if** $W_{\text{total}} = 0$ **else** S_w/W_{total} $\triangleright$ Fallback to longest.

The second algorithm performs a weighted average, as shown in [Alg. 3](#). We bias these towards the shorter-range arrays to maximize accuracy. We formulate a linear weight curve for each array with y-intercept $1/s_i$ and x-intercept $C \cdot s_i$. We set C to 2 as arrays are known to be mostly effective up to $2 \times$ their dwell time, and compute s_i proportional to the array dwell time τ_i . We define two operational modes, linear and sqrt, setting $s_i = \tau_i$ and $s_i = \sqrt{\tau_i}$, respectively. Sqrt mode assigns a higher weight to short-range arrays at the start. The weight for each estimate t_i is calculated from the linear weight curve; the weighted average determines the final time estimate. The longest dwell-time array is the fallback if all weights are zero.

5 Evaluation Methodology

We evaluate our design through a combination of real-world experiments and simulation. Our simulations were informed and validated by the real-world data we gathered.

5.1 Real-World MTJ Experiments

To confirm that physical MTJs exhibit the stochastic characteristics required for FLINT's timekeeping approach, we characterize 21 devices fabricated using the same stack composition and layer thicknesses as reported in [8]. The measured devices are voltage-controlled MTJs (V-MTJs), in which the energy barrier (E_b) can be tuned via an applied DC voltage through the voltage-controlled magnetic anisotropy (VCMA) effect. This allows a single set of physical devices to be characterized across multiple target dwell

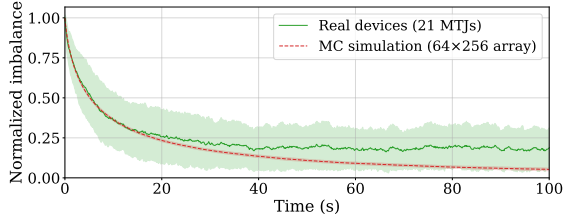


Figure 10: Sim validation: normalized imbalance for 21 fabricated MTJs (green, $\pm 1\sigma$) vs. Monte Carlo sim of 64×256 array (red dashed).

time regimes without requiring separate fabrication runs, emulating the behavior of different fixed- E_b devices. Although the final system operates using STT for switching and reset, VCMA is used here purely as a characterization tool to modulate E_b ; it does not alter the underlying thermal switching dynamics measured.

We configured each device for two operating points, $\tau = 10$ s and 25 ms, and recorded the time between switching events (10 s is generally useful; 25 ms targets high resolution). From the measured switching event sequence, we computed per-device τ and applied the Néel-Arrhenius relation to extract the corresponding E_b . In practice, MTJs with τ spanning milliseconds to hundreds of seconds can be created by adjusting the free layer volume, anisotropy, or material stack during fabrication [48]. This corresponds to E_b on the order of $10\text{--}40k_bT$, making our target operating points representative of deployable FLINT arrays.

5.2 Array Simulations

We evaluate FLINT hierarchically at the array level. We first study a single MTJ array to characterize fundamental accuracy–range trade-offs and sensitivity to design parameters. We then extend the analysis to multi-array configurations, where heterogeneous arrays are composed to improve timekeeping range and robustness.

We model FLINT using Monte Carlo simulations that capture the stochastic switching behavior of MTJs and their aggregate effect on time estimation. Device characteristics are derived from validated hardware models [32, 33] and experimental measurements (e.g., E_b , TMR). Fig. 10 confirms that the simulator accurately reproduces measured device behavior (validation methodology in App. A).

We perform sensitivity analyses over key design parameters, sweeping each parameter independently and reporting results as geometric means over 100 trials to ensure statistical robustness. Since ADC sampling latency ($\approx 1 \mu\text{s}$ [63]) is negligible relative to MTJ dwell times, it is omitted.

Single-Array Experiments. To quantify energy consumption and guide the design of our control hardware, we use SPECTRE-based hardware simulations [32, 33]. We then evaluate FLINT using sensitivity analyses across key design dimensions, including sensing parameters (ADC precision, noise, and measurement voltage), array configuration (size and aspect ratio), calibration parameters (run count and table size), device variability (E_b variation), and environmental factors (temperature). Default values are listed in Table 1, and we set $E_b = 23 kT$ ($\tau = 9.74$ s). Unless otherwise stated, parameters are chosen based on hardware measurements (e.g., E_b ,

Table 1: Device and System Configurations.

Parameter	Default Value
ADC Bits	12
ADC Noise	1 ADC Step
MTJ Array Size	64×256
Calibration Run Count	20
Calibration Table Size	128 entries/array
E_b std. dev.	6.3%
Temperature	293 K
V_{meas}	0.052 V per MTJ (VDD = 3.3 V)
TMR	124.49%, $\sigma = 7.81\%$
R_p	52.66 k Ω , $\sigma = 3.73$ k Ω
R_{shunt}	27.5 k Ω
Fusion Method	LinearSqrtDecay

TMR, R_p) or representative MCU capabilities (e.g., 12-bit ADC [63]). Detailed impacts of these parameters are discussed in Sec. 6.

Multiple-Array Experiments. We evaluate a multi-array configuration consisting of four MTJ arrays with heterogeneous energy barriers of 23/24/25/26 kT ($\tau = 9.74/26.5/72.0/196$ s) and identical default parameters (Table 1). We focus on two key factors: robustness to uncalibrated temperature variation and the effectiveness of different fusion methods. We compare our proposed fusion strategies (Sec. 4.4) against a simple averaging baseline representative of lightweight implementations. We also attempted to add HARC-Lite [20], but could not reproduce the results using the pseudocode.

Metrics. We report several metrics: (i) percent error over time, (ii) absolute error due to E_b variation for short durations ([0, 5] s), (iii) time to 10% error, the effective timekeeping range, (iv) energy overhead, and (v) latency.

5.3 Comparison with State-of-the-Art

We compare FLINT against prior timekeeping approaches using two complementary evaluations. First, we analyze system-level characteristics, including measurement range, energy consumption, startup latency, and resolution. Second, we evaluate time-estimation accuracy using four real energy-harvesting traces from kinetic and solar sources [35] to capture realistic operating conditions.

For system-level comparison, we use three capacitors of 22 nF, 220 nF, and 2.2 μF for CUSTARD [40], HARC [20], and CHRT [23]. We assume 0805 capacitors to estimate the size. We configure three arrays in FLINT with dwell times to achieve the same range as the other systems. The calibration table contains 1024 entries to ensure comparability with CHRT [23]. All FLINT results are aggregated across 100 runs. For trace-based evaluation of CHRT and FLINT, we use four real energy-harvesting traces and evaluate performance under different energy buffer sizes of 10 μF , 100 μF , and 1 mF, reflecting a range of intermittent operating conditions.

5.4 Long-Term Aging Comparison

We compare FLINT against an ideal capacitor-based timekeeper based on the C3 (2.2 μF) configuration from Sec. 5.3, which provides the longest range (91.3 s) among the evaluated capacitor-based designs. We model capacitance degradation (following [16]) as $C(d) = C_0(1 - \alpha d/365)$, where d is the deployment day and α is the annual degradation rate. We evaluate conservative ($\alpha = 0.20$,

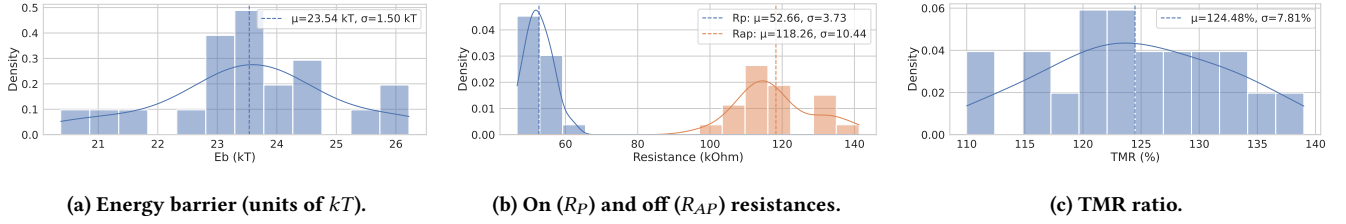


Figure 11: MTJ parameter distributions for devices with a 10 s dwell time.

20 %/yr) and worst-case ($\alpha = 0.50$, 50 %/yr) aging scenarios. To isolate the effect of aging, we attribute all capacitor clock error solely to capacitance degradation and exclude measurement, quantization, and leakage effects, intentionally favoring the capacitor-based approach. As capacitance decreases, the system range scales with $C(d)$, and measured time is overestimated by $C_0/C(d)$. In contrast, FLINT's MTJ energy barriers (E_b) are unaffected by charge-discharge cycling, and its timekeeping behavior therefore remains unchanged throughout the deployment. To quantify application-level impact, we use the FLINT configuration from Sec. 5.3 and evaluate a representative sensing task scheduled every 60 s (a common sensing interval), counting re-sample mistakes over a simulated day.

6 Results

We evaluate FLINT through array-level simulations, energy analysis, comparison with SOTA, aging analysis, and case studies. Results are presented in the following sections.

6.1 Device Characterization

The resistance and TMR characteristics of the devices are shown in Fig. 11. The R_P and R_{AP} distributions (Fig. 11b) confirm well-separated resistance states with low device-to-device variability, and a mean TMR of 124.45% ($\sigma = 7.81\%$) (Fig. 11c) provides sufficient contrast for reliable ADC-based resistance measurement.

The extracted E_b distribution at both operating points is shown in Fig. 11a and Fig. A2. At the 10 s target, devices have a mean $E_b = 23.5 kT$ ($\sigma = 1.50$); at 25 ms, the mean shifts to $E_b = 17.0 kT$ ($\sigma = 1.17$). Both distributions are well-described by a normal centered near the target E_b , confirming that device-to-device variation is bounded and predictable across dwell-time regimes. These measurements are the basis for the simulation parameters in Sec. 5.2.

6.2 Single Array Trade-off Analysis

We study the impact of key design parameters on timekeeping accuracy and identify configurations that achieve ≥ 40 s of timekeeping within 10% mean error. Further analysis is also provided in App. C.

Impact of ADC Noise. We evaluate the impact of ADC noise on timekeeping accuracy, as shown in Fig. 12. Noise is injected at the MTJ output in multiples of the ADC quantization step, as only noise exceeding this resolution affects the measured voltage. We observe that the system remains robust up to 2.4mV (3 quantization steps on a 12-bit ADC at 3.3V VDD), beyond which the timekeeping range degrades by nearly 50%. This is explained by the intrinsic range separation between MTJ states: even under worst-case variation,

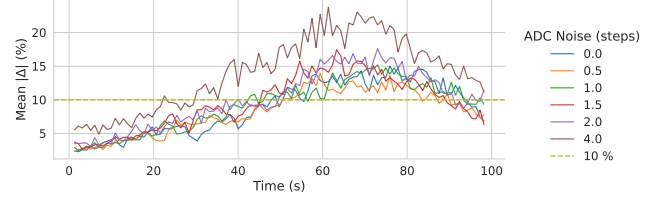


Figure 12: Timekeeper error with varying ADC noise.

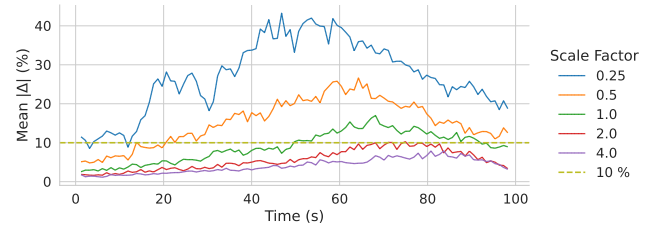


Figure 13: Timekeeper error with varying array scales.

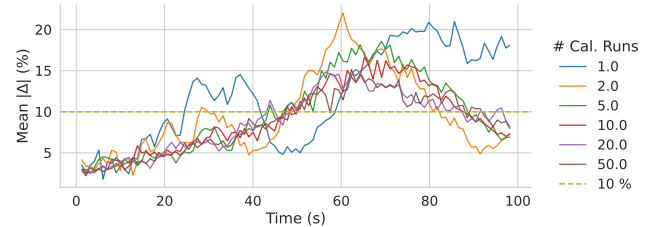


Figure 14: Timekeeper error with varying calibration runs.

the R_P - R_{AP} voltage difference remains $6\times$ larger (see Fig. 11b) than the ADC quantization step, providing sufficient noise margin.

Impact of Array Size. In Fig. 13, we evaluate the impact of array size by scaling a 64×64 MTJ array. Increasing the number of MTJs improves precision and timekeeping accuracy. However, larger arrays require higher reset voltages: for example, a 64×64 array may require ~ 10 V per row, while a 256×256 array would increase this requirement to ~ 40 V, which may be impractical for intermittent battery-free systems. This highlights a fundamental trade-off: array size can be increased to improve accuracy, but is ultimately constrained by the reset voltage supported by the system.

Impact of Calibration. We evaluate the impact of calibration effort by varying both the number of calibration runs and the size

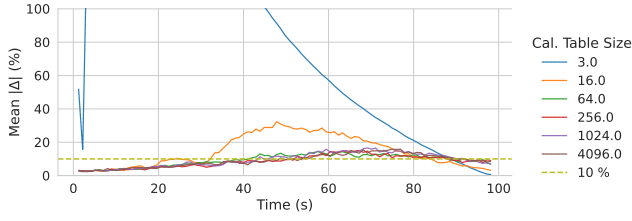
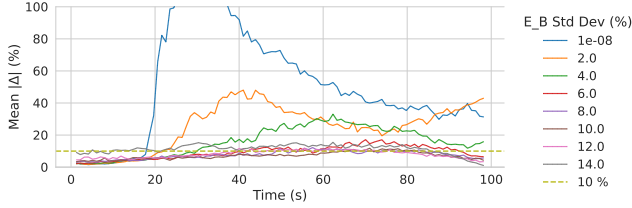
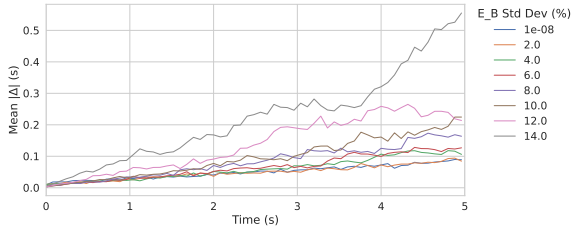


Figure 15: Impact of varying calibration table size.

Figure 16: Impact of varying MTJ E_b standard deviations.Figure 17: Absolute error with varying MTJ E_b standard deviations, in the $[0, 5]$ second interval.

of the calibration table, as shown in Fig. 14 and Fig. 15, respectively. Increasing calibration runs reduces noise in the learned mapping, while larger tables improve resolution through finer-grained resistance-to-time mapping. We observe diminishing returns beyond moderate configurations: 5–10 calibration runs are sufficient, and at least 64 table entries are required to achieve a 40 s timekeeping range within 10% error. We select a 128-entry table (1 KiB) as our default. While increasing these parameters improves accuracy, it also introduces overhead in calibration time, memory, and latency, similar to prior approaches [20, 23].

Impact of E_b Variability. We study the impact of process-induced variation in the energy barrier E_b by varying its standard deviation while keeping the mean constant (Fig. 16). Low variability yields minimal error at short durations but limits the timekeeping range, as most MTJs switch at similar times. In contrast, higher variability spreads switching events over time, slightly increasing error but sustaining sub-10% error for longer durations. We find that a 4–8% standard deviation—consistent with typical MTJ fabrication—provides an effective operating point, balancing accuracy and range. This highlights a key insight: FLINT not only tolerates process variation, but benefits from it under realistic conditions.

Impact of Array Squareness. We define the array squareness as the ratio of the number of columns to the number of rows. In

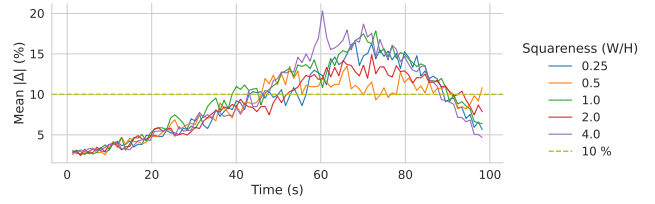


Figure 18: Timekeeper error vs MTJ array squareness

Fig. 18, we plot the error as a function of squareness. We find that squareness does not impact the error or the timekeeping range. Thus, when reset voltage is a constraint, the array width may be reduced with very minimal impact.

Impact of Temperature. We evaluate the impact of runtime temperature on timekeeping accuracy in Fig. 19a. Without correction, temperature variation degrades accuracy, especially at higher temperatures. Adding runtime temperature correction (Sec. 4.3) keeps error within 10% in a ± 20 K range for a 40 s interval, when physically possible (dwell times decrease at higher temperatures, limiting range). We further compare against calibration at different temperatures in Fig. 19b, where runtime and calibration temperatures are matched, achieving a similar 40 s range with slightly lower error. These results show that lightweight runtime correction, enabled by widely available on-chip sensors [75] or calibration, is sufficient to maintain accuracy under realistic temperature variation.

6.3 Multiple arrays

A single array supports up to 40 s timekeeping within 10% error. To extend this range, we compose four MTJ arrays and evaluate their combined behavior.

MTJ dwell times and corresponding energy barriers follow stochastic distributions, which differ across arrays. Naïve averaging fails to capture these dynamics, as it assumes a stationary and identical distribution across arrays. As shown in Fig. 20a, this results in high error, rendering the approach unreliable. In contrast, fusion methods that assign adaptive weights to observations—particularly those that prioritize more recent data—better capture the evolving system dynamics and significantly improve stability. All three methods achieve a 938 s range within 10% error; we use Linear-Decay Sqrt as the default, as it achieves the lowest error on average.

We evaluate robustness under a ± 20 K temperature variation in Fig. 20b. Runtime temperature correction using 4 arrays also achieves a 938 s range under 10% error when physically possible.

6.4 Energy, Time, and Space Overheads

Energy: Using a Cadence MTJ simulator [32, 33], we determine that a 50 ns, 35 μ A pulse reliably resets a single MTJ. Thus, in the worst case (AP state, $R = 118.26$ k Ω), the energy needed to reset a single MTJ is $E_{reset} = I^2 \cdot R \cdot t = 7.24$ pJ. Thus, with four 64×256 arrays (65 536 devices) the total required energy is $\widehat{E}_{reset} = 474.7$ nJ.

Then, to measure the MTJs, there are three energy costs: array/shunt resistor, ADC, and software. The energy dissipated in the array and shunt resistor in the worst case (P state, $R_{array} = 13.2$ k Ω) can be easily calculated. Assuming an ADC sample time of 1 μ s [63], the total energy comes to $\widehat{E}_{meas} = 4.28$ nJ. The ADC energy is 87 nJ

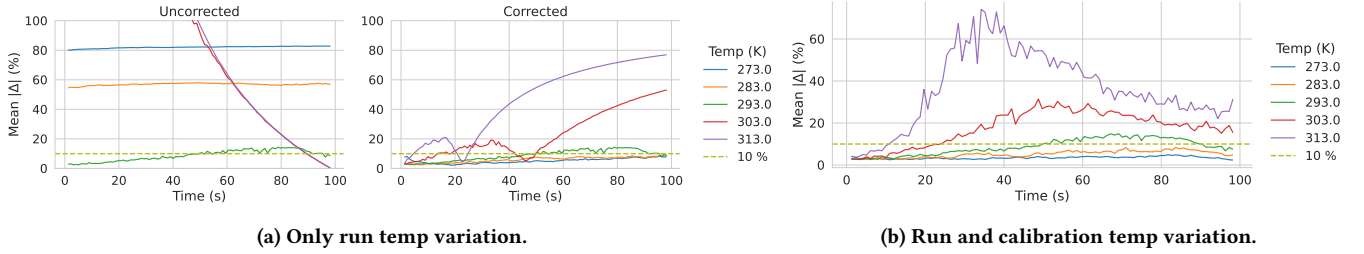


Figure 19: Timekeeper error with varying temperatures, at runtime and (optionally) calibration time.

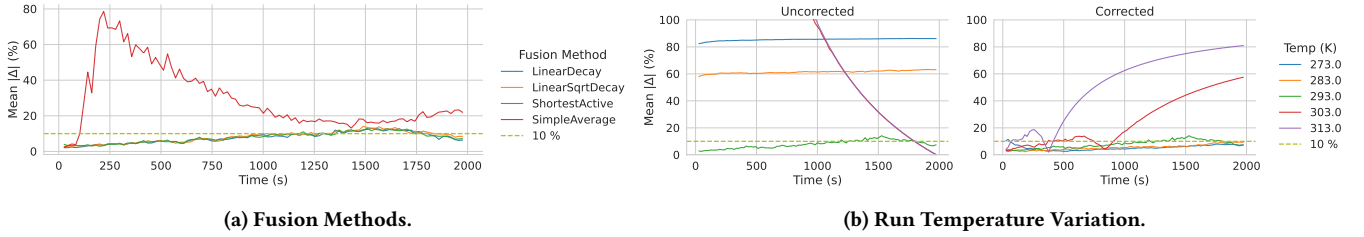


Figure 20: Multiple array timekeeper error with varying fusion methods and temperature variations.

per channel (experimentally measured on a MSP430FR5994 [63]) for a total of $E_{ADC} = 348$ nJ across the four arrays.

To estimate software energy, we utilized `llvm-mca` [51]. Since `llvm-mca` does not support the MSP430 ISA, we target an ARM Cortex-A15 [6] with `clang++` (commit 72df1fc6); the Cortex-A15 is higher-power than typical batteryless MCUs, giving us a conservative upper bound on software energy. We determine that the software time estimation code takes 401 cycles to complete. Using the energy estimations in [64], total software energy is estimated as $E_{sw} = 207.74$ nJ. Thus, total FLINT system energy use is

$$E_{total} = \widehat{E}_{reset} + \widehat{E}_{meas} + E_{ADC} + E_{sw} = 1034.72 \text{ nJ} \quad (4)$$

Time: At the MSP430’s minimum 6 MHz, 401 cycles take 66.83 μ s, a conservative estimate; adding four 1 μ s ADC samples [63] gives 70.83 μ s total startup overhead. Shutdown time overhead is the reset pulse length: 50 μ s.

Memory: Each array has a calibration table with 128 entries of two 4-byte floats each, totaling 1024 B of space. Thus, four tables need 4 KiB, which easily fits in the ≈ 512 KiB of non-volatile memory found in common MCUs [30, 63]. If we only store time *values*, and compute the `LinSpace` (Alg. 1) every startup, this can be reduced to 2 KiB for the tables, and an additional 12 B for `LinSpace` parameters, totaling 2060 B. However, this incurs additional compute overheads.

Size: An MTJ is 200 nm in diameter, including spacing to prevent magnetic coupling [74]. Thus, a 64×256 array requires $655.36 \mu\text{m}^2$. Within 0.1 mm^2 , we could fit 100 arrays!

6.5 Comparison of FLINT with the SOTA

We summarize a comparison of FLINT against prior timekeepers in Table 2, including an Ambiq AM1805 RTC [4]. The E_{timer} row reflects the cost to charge or reset the timing element only; ADC and software costs are itemized separately and apply equally to all approaches (Sec. 6.4). FLINT significantly extends the maximum

Table 2: Comparison of FLINT with SOTA timekeepers.

	C1	C2	C3	CHRT	HARC	AM1805	FLINT
Range (s)	0.91	9.13	91.3	102	91.33	110	273.6
I_{static} (nA)	30	30	30	958	0	55	0
E_{timer} (μ J)	0.098	0.98	9.85	10.94	10.94	0	0.356
E_{ADC} (nJ)	87	87	87	261	261	0	261
$t_{startup}$ (ms)	5.4	54.6	546.7	607.24	607.24	900	0.071
Resolution (s)	0.01	0.1	1	0.01-1	0.01-1	0.01	0.005-1
NVM Use (B)	8192	8192	8192	24576	11001	N/A	24576
Size (mm^2)	2.5	2.5	2.5	7.5	7.5	9	0.002

timekeeping range to 273.6 s, achieving $3.00\times$ and $2.68\times$ longer durations than HARC and CHRT, respectively. At the same time, it achieves time resolution as low as 5 ms, while maintaining a worst-case resolution of 1 s—equivalent to prior approaches. When the range is limited to 100 s, the worst-case resolution can be improved to 0.19 s. Unlike prior work, FLINT needs no dedicated capacitors or active circuitry for off-time measurement, eliminating startup latency and active current. The only required operation is one ADC measurement per array after each power interruption, similar to prior approaches, which require one measurement per capacitor.

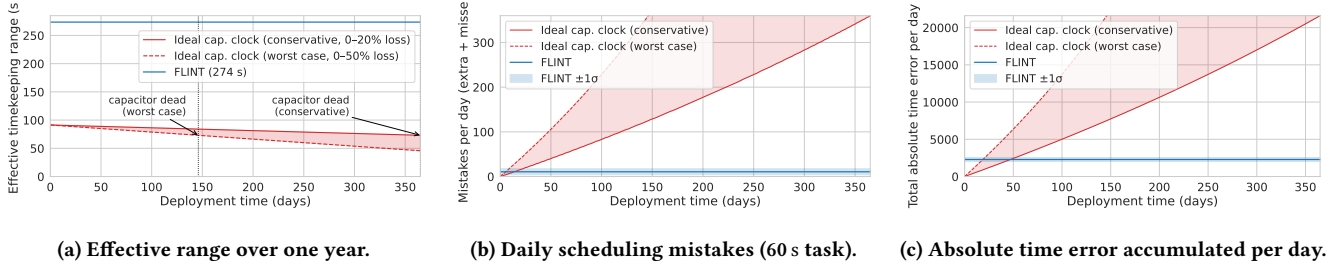
We further evaluate time-estimation accuracy using real energy-harvesting traces (Table 3). CHRT is unable to reliably track time under long and highly variable off-time conditions, as its charging-time-based estimation cannot cover extended interruptions, whereas FLINT maintains accurate estimation. Both methods achieve similar accuracy under stable conditions, but FLINT significantly extends the operating range and enables more applications with its lower energy use, while preserving estimation quality.

6.6 Long-Term Reliability Under Aging

The comparison above assumes a fresh capacitor; Fig. 21 shows how aging erodes that baseline over one year.

Table 3: Estimation error (%) for FLINT vs. CHRT. CHRT has startup time before execution. DNF = capacitor could not charge.

	Trace A			Trace B			Trace C			Trace D		
Sys. Cap. Size	10 μ F	100 μ F	1 mF	10 μ F	100 μ F	1 mF	10 μ F	100 μ F	1 mF	10 μ F	100 μ F	1 mF
Avg. Off Time (s)	1.55	15.22	190.54	2.86	12.62	168.22	0.47	4.35	45.24	5.03	51.09	DNF
CHRT	2.25%	Failed	Failed	0.83%	3.14%	Failed	0.71%	2.94%	Failed	3.32%	Failed	—
FLINT (%)	1.69%	2.52%	3.09%	1.60%	1.34%	1.32%	1.64%	1.07%	1.49%	2.05%	2.40%	—

**Figure 21: FLINT vs. ideal capacitor clock over one year. Bands: conservative (0–20%/yr) to worst-case (0–50%/yr) degradation [16].**

Range. FLINT holds 273.6 s for the device’s life, while the ideal capacitor clock falls from 91.3 s to 73.0 s (conservative) or 45.7 s (worst case) within a year, crossing a 20% loss “capacitor dead” threshold at day 365 and day 146 respectively. FLINT’s 3.0 $\times$ range advantage at deployment grows monotonically thereafter.

Accuracy and scheduling. The accuracy cost is sharper than the range cost. Using the re-sample mistake metric defined in Sec. 5.4 (fixed 60 s scheduled task), the ideal capacitor clock’s aging bias has a fixed sign—it always reads high and fires early—so the error does not average out but accumulates across the thousands of cycles in a day (Fig. 21b). Averaged over a year, the ideal capacitor clock incurs 60 889 redundant samples (conservative) to 203 659 (worst case), versus roughly 1992 redundant and 1889 missed for FLINT, whose errors are zero-mean and do not accumulate. This is a 16 $\times$ to 52 $\times$ reduction. The same one-directional bias inflates the total mistimed duration: summed over a day, the ideal capacitor clock’s absolute accumulated time error grows with age and overtakes FLINT’s flat error within roughly the first two months of deployment (Fig. 21c).

6.7 Application Case Studies

We evaluate FLINT on five representative intermittent computing applications, demonstrating its versatility across timing regimes that capacitor clocks either handle poorly or cannot reach at all.

- **A1 Position logging** [44, 55, 62]: a batteryless GPS tag fires periodic fixes to log wildlife trajectories; timing accuracy is critical to avoid wasted 9 mJ GPS acquisitions and stale position data.
- **A2 Structural health monitoring** [1]: a batteryless sensor on a bridge samples vibration at regular intervals; capacitor aging introduces cumulative drift that biases the collected data.
- **A3 Cold-chain monitoring** [57]: a batteryless tag accumulates unsafe-temperature exposure; directional aging bias of capacitors leads to wrong safety verdicts on shipments.
- **D4 Overnight solar gap**: a solar-powered node must estimate time elapsed at dawn to decide whether a buffered reading is still fresh; the 12 h gap exceeds any practical capacitor.

- **D5 Far-future wakeup**: a wildlife tag must schedule a weekly radio transmission; a 7 d off-time is impossible to measure with any capacitor.

The first three (A1–A3) prioritize *measurement accuracy*: off-times fall within the capacitor clock’s range, enabling direct comparison against an aging capacitor baseline (C3 from Sec. 5.4). Off-times are drawn from a shifted-binomial model $t = b + s \cdot \text{Bin}(n, 0.5)$ with means below the C3 baseline’s ~ 91 s ceiling, isolating aging bias from range effects. The next two (D4–D5) prioritize *measurement range*: the required off-times exceed the practical capacitor ceiling, so we report FLINT error at the endpoint alongside the capacitor energy and area that *would* be required. FLINT uses the default multi-array configuration from Sec. 5.2; E_b overrides are in Table 4. Each application defines a $\pm 10\%$ tolerance on its target interval T^* ; mistake rates are the arithmetic mean across 20 device realizations. **Accuracy analysis** (Table 5). Capacitor aging introduces an ever-increasing timing bias. The capacitor makes zero mistakes until this bias exceeds 10%—day 183 ($\alpha=0.2$) or day 74 ($\alpha=0.5$)—after which decisions become systematically wrong. FLINT’s errors carry no consistent bias and do not accumulate with deployment age.

A1: FLINT makes zero early-fire or stale-data mistakes across all 20 trials. The aging capacitor fires every GPS fix more than 1 s early from day 74–183 onward, accumulating 1.86–3.71 million premature 9 mJ fixes per year (35 kJ/yr wasted [55]).

A2: FLINT has zero mis-timed samples across all 20 device trials. The aging capacitor mis-schedules 20 700–41 200 samples per year.

A3: At the default 1.1B condemn threshold, FLINT makes zero false condemnations; 1/20 device realizations miss a spoiled shipment (18/yr). Lowering the threshold to B eliminates all missed condemnments with zero false condemnments—an application-level tuning with no hardware change. The aging capacitor falsely condemns safe shipments from day 100–249 onward (117–266 wrong verdicts per year)—a one-sided, growing failure mode absent in FLINT.

Range Analysis (Table 6, Fig. 22). FLINT tracks D4 at 2.2 % and D5 at 2.3 % geomean error. A capacitor tops out near 91 s; reaching

Table 4: Case study configurations. Note $t_{off} = b + s \cdot \text{Bin}(n, 0.5)$, and $\bar{t}_{off}$ = mean off time.

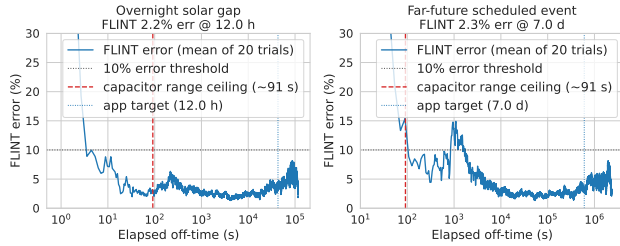
ID	Application	b	s	n	$\bar{t}_{off}$	Mistake definition	E_b (kT)
A1	Position logging [44, 55, 62]	1.0 s	0.2 s	30	4 s	Early fire: true elapsed < 9 s; stale data: true age > 66 s	23–26
A2	Structural health monitoring [1]	10 s	1 s	50	35 s	Sample outside [810 s, 990 s] ($\pm 10\%$ of $T^* = 900$ s)	23–26
A3	Cold-chain monitoring [57]	10 s	1 s	30	25 s	False condemn ($\hat{E} > 1.1B$, safe) or missed condemn ($\hat{E} \leq 1.1B$, spoiled)	23–26
D4	Overnight solar gap	target: ≈ 12 h				Geomean % error (no cap baseline)	28–31
D5	Far-future wakeup	target: ≈ 7 d				Geomean % error (no cap baseline)	31–34

Table 5: Accuracy case study mistakes per year.

Application	FLINT/yr	Opt. cap/yr (con.-wst.)
Position logging	0	1.86M – 3.71M
Struct. health	0	20.7K – 41.2K
Cold-chain	18 (0 tuned)	117–266

Table 6: Duration case study results.

Application	Range	FLINT err.	Cap. energy	Cap. size
Overnight solar gap	12 h	2.2 %	4.7 mJ	1.0 mF
Far-future event	7 d	2.3 %	65.2 mJ	14.6 mF

**Figure 22: FLINT error in duration case studies, reaching hours-to-days ranges at $\sim 2\%$ err—impossible for a capacitor.**

these ranges would require 4.7 mJ–65.2 mJ per cycle and 1.0 mF–14.6 mF of bulk capacitance for D4 and D5, respectively.

6.8 Discussion and Limitations

While FLINT’s physics-based approach avoids the structural failure modes of capacitor clocks, several practical challenges remain.

High reset voltage: Reset voltage scales with array width due to the high resistance of MTJs. This can be mitigated by reducing device resistance (e.g., via area or tunnel barrier engineering) without affecting decay dynamics, changing squareness, or by redesigning the reset circuitry to operate on smaller sub-arrays. Alternative MTJ technologies (e.g., SOT or hybrid SOT-STT [45]) may further reduce switching requirements.

Calibration overhead: Calibration requires ADC sampling over the full decay interval (up to ~ 1500 s sampled every 5 ms). While this incurs time and energy overhead, it is a one-time process at setup and comparable to capacitor-based approaches. Future work may explore standardized calibration to reduce this cost.

Device lifetime: MTJ endurance ($> 10^{12}$ switching events) far exceeds timekeeping needs, and aging does not measurably alter

switching behavior [5, 19]. The reset/measure cycle is the only wear path; at any plausible cycle rate the device outlives deployment.

Fabrication challenges: Scaling MTJ fabrication introduces variability and integration challenges. However, ongoing advances in MRAM technology continue to improve device uniformity and manufacturing, supporting the feasibility of large-scale deployment.

Evaluation scope: Exhaustive experimental design-space exploration is impractical: each array configuration (width, height, size, E_b , R_p , TMR) requires a custom CMOS tape-out. Our design-space study therefore relies on simulation validated against fabricated devices (App. A). Area figures are estimates: capacitor sizes assume 0805 packages; FLINT’s $< 0.1 \text{ mm}^2$ is a pessimistic upper bound.

7 Related Works

7.1 Time-Sensitive Intermittent Computing

Several software runtimes support time-sensitive execution in intermittent systems [29, 38, 46, 67, 68, 72], maintaining data freshness by restarting tasks when collected data becomes stale. Intermittent real-time scheduling approaches [42, 43, 53] similarly incorporate temporal constraints into task execution across power cycles. Both depend on accurate elapsed-time estimates across power failures, making persistent timekeeping a fundamental requirement.

7.2 Timekeeping Approaches

Prior work has explored time estimation without dedicated hardware by leveraging existing system components. TARDIS [39] estimates elapsed time from SRAM remanence, while oscillator-based approaches infer duration from stabilization behavior [3]. SQUID [73] predicts off-time by modeling capacitor charging during active periods. More recently, BIONIC [69] estimates ambient power from capacitor charging times and predicts future charging delays, enabling adaptation to changing energy conditions without hardware modifications. However, its accuracy ultimately depends on the stability of harvested-power patterns and the validity of the underlying prediction model. Overall, these approaches either support only limited time ranges or rely on environment-dependent assumptions that can reduce robustness in changing conditions.

Capacitor-discharge techniques remain the dominant approach for long-range timekeeping. CUSTARD [39] uses a single capacitor, exposing a fundamental trade-off between startup cost and measurable duration: small capacitors charge quickly but support only short-range measurements, whereas larger capacitors extend range at the expense of energy overhead and startup latency. CHRT [23, 70], HARC [20], and federated capacitor designs [7] improve accuracy and range through multiple storage elements, but still inherit

Table 7: Comparison of timekeeping approaches across key design axes. Colors: Fully satisfies | Partially satisfies | Does not satisfy.

System	Category	Time Source	Aging Sensitivity	Environmental Robustness	Resolution	Max Range	Energy vs. Range	Startup Overhead	Reconfig Required?
AM1805 [4]	HW	RTC	Medium (BCA)	Medium (BCA)	High	High	Linear	Very High	Yes
TARDIS [39]	SW	SRAM	None	Low (Temperature)	Low	Low	Const (Low Range)	Low	No
Osc.-based [3]	SW	Osc.	None	Low	High	Very Low	Const (Low Range)	Low	No
CUSTARD [39]	HW	Cap.	Cap. Drift	Low	High	Low	Linear	Scales w/ Cap.	Yes
SQUID [73]	SW	HE	Model Drift	Very Low (PF)	High	Low	Const (Low Range)	Low	No
CHRT [23]	HW	Cap.	Cap. Drift	Low	High	High	Linear	Scales w/ Cap.	Yes
HARC [20]	HW	Cap.	Cap. Drift	Low	High	High	Linear	Scales w/ Cap.	Yes
Fed.-Cap. [7]	HW	Cap.	Cap. Drift	Low	High	High	Linear	Scales w/ Cap.	Yes
BIONIC [69]	HW/SW	Cap./HE	Cap. Drift	Very Low (PF)	High	High	Linear (w/ SR)	Low	No
FLINT (this work)	HW/SW	MTJ	None	High	High	High	Constant	Low	No

SRAM = SRAM remanence Osc. = oscillator stabilization Cap. = capacitor discharge HE = harvested-energy modeling
 PF = power fluctuations BCA = backup capacitor aging ET = environmental temperature SR = sampling rate

the same dependency on capacitor sizing. Consequently, measurement range remains coupled to energy consumption, area, and startup cost. Furthermore, because time estimation relies directly on capacitor behavior, accuracy is affected by capacitor aging and environmental variation over the deployment lifetime (Sec. 6.6).

Table 7 summarizes representative timekeeping approaches across key design axes. SRAM-remnance and oscillator-based techniques offer aging stability and low startup overhead, but are fundamentally limited in range. RTC-based solutions provide long-range, high-resolution timekeeping; however, like other capacitor-backed designs, they require deployment-time capacitor sizing based on expected harvesting conditions, potentially necessitating reconfiguration when those conditions change. Harvested-energy modeling approaches avoid explicit capacitor characterization but remain highly dependent on environmental conditions and often require retraining or recalibration. In contrast, FLINT achieves long-range, high-resolution, and environmentally robust timekeeping while maintaining constant energy cost, low startup overhead, and operation without hardware reconfiguration.

7.3 Other MTJ Applications

Outside of timekeeping, MTJs are most widely deployed as the storage element in STT-MRAM, where resistance states retain data non-volatily with high endurance and radiation tolerance [41, 54, 65]. They also serve as entropy sources: a fixed current across an STT-MTJ [61] or fixed voltage across a V-MTJ [60] drives 50% switching probability, producing the random bitstreams required by p-bits [13, 50] and p-dits [25, 26, 28] used in probabilistic computing to solve combinatorial optimization problems [11, 27]. Additionally, the sensitivity of the free layer to external magnetic fields enables use as magnetic field sensors [49].

8 Conclusion

We present FLINT, the first timekeeper to read elapsed time from the physics of broken memory rather than from stored energy. Because an MTJ array’s decay timescale is set by device geometry, FLINT’s energy is independent of the duration it measures and its accuracy does not drift as the device ages. These are the two failure modes that limit capacitor clocks in long-lived deployments. FLINT tracks over 15 minutes of off-time within 10% error while consuming 1.03 μJ , occupying under 0.1 mm^2 , and adding 50 μs of reset latency

and 70.83 μs to estimate time, and it extends to longer ranges at no added cost. Because idle arrays consume no power and a single chip can host many more arrays than any deployment needs, the same hardware covers short-range high-resolution tasks and long-range low-resolution ones without modification—a true “one size fits all” timekeeper, in contrast to capacitor clocks whose storage element must be sized at deployment time. Lightweight calibration and temperature correction hold this accuracy across realistic operating conditions, and over a one-year deployment FLINT makes 16–52 $\times$ fewer scheduling errors than an aging capacitor clock.

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References

- [1] Mikhail Afanasov et al. 2020. Battery-less zero-maintenance embedded sensing at the mithraeum of circus maximus. In *Proceedings of the 18th Conference on Embedded Networked Sensor Systems*, 368–381.
- [2] Saad Ahmed, Naveed Anwar Bhatti, Muhammad Hamad Alizai, Junaid Haroon Siddiqui, and Luca Mottola. 2019. Efficient intermittent computing with differential checkpointing. In *Proceedings of the 20th ACM SIGPLAN/SIGBED International Conference on Languages, Compilers, and Tools for Embedded Systems*, 70–81.
- [3] Arwa Alsubhi, Nicole Tobias, Simeon Babatunde, and Jacob Sorber. 2020. Can crystal oscillators keep time without power? In *Proceedings of the 8th International Workshop on Energy Harvesting and Energy-Neutral Sensing Systems*, 84–85.

- [4] Ambiq Micro, Inc. 2022. Artiasie am18x5 real-time clock datasheet. <https://ambiq.com/wp-content/uploads/2020/10/Artiasie-AM18X5-RTC-Datasheet.pdf>. Accessed: 2026-04-01. (2022).
- [5] Dmytro Apalkov, Bernard Dieny, and Jon M Slaughter. 2016. Magnetoresistive random access memory. *Proceedings of the IEEE*, 104, 10, 1796–1830.
- [6] ARM. 2013. Cortex-A15 MPCore Processor Technical Reference Manual r4p0. ARM Developer. (June 24, 2013). Retrieved Apr. 15, 2026 from <https://developer.arm.com/documentation/ddi0438/i/?lang=en>.
- [7] Alberto Rodriguez Arreola, Theodoros D Verykios, Marco A Gurrola Navarro, and Carlos F Calvillo Cortes. 2022. Federated time persistency in intermittently powered iot systems. *Journal of Systems Architecture*, 130, 102667.
- [8] Jordan Athas, Christian Duffee, Thomas Neuner, Noraica Davila Melendez, Jordan A Katine, and Pedram Khalili Amiri. 2025. Statistical study of large voltage-controlled magnetic anisotropy in 3x-nm perpendicular magnetic tunnel junctions. *IEEE Electron Device Letters*.
- [9] Abu Bakar, Rishabh Goel, Jasper de Winkel, Jason Huang, Saad Ahmed, Bashima Islam, Przemyslaw Pawelczak, Kasim Sinan Yildirim, and Josiah Hester. 2022. Protean: an energy-efficient and heterogeneous platform for adaptive and hardware-accelerated battery-free computing. In *Proceedings of the 20th ACM Conference on Embedded Networked Sensor Systems*, 207–221.
- [10] Domenico Balsamo, Alex S Weddell, Anup Das, Alberto Rodriguez Arreola, Davide Brunelli, Bashir M Al-Hashimi, Geoff V Merrett, and Luca Benini. 2016. Hibernus++: a self-calibrating and adaptive system for transiently-powered embedded devices. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 35, 12, 1968–1980.
- [11] William A Borders, Ahmed Z Pervazi, Shunsuke Fukami, Kerem Y Camsari, Hideo Ohno, and Supriyo Datta. 2019. Integer factorization using stochastic magnetic tunnel junctions. *Nature*, 573, 7774, 390–393.
- [12] Adriano Branco, Luca Mottola, Muhammad Hamad Alizai, and Junaid Haroon Siddiqui. 2019. Intermittent asynchronous peripheral operations. In *Proceedings of the 17th Conference on Embedded Networked Sensor Systems*, 55–67.
- [13] Kerem Yunus Camsari, Rafatul Faria, Brian M Sutton, and Supriyo Datta. 2017. Stochastic p-bits for invertible logic. *Physical Review X*, 7, 3, 031014.
- [14] Nilotpal Chakravarti. 1989. Isotonic Median Regression: A Linear Programming Approach. *Mathematics of Operations Research*, 14, 2, (May 1989), 303–308. doi:10.1287/moor.14.2.303.
- [15] Zhuo Cheng, Bradley Denby, Kyle McCleary, and Brandon Lucia. 2024. Eagleeye: nanosatellite constellation design for high-coverage, high-resolution sensing. In *Proceedings of the 29th ACM International Conference on Architectural Support for Programming Languages and Operating Systems, Volume 1*, 117–132.
- [16] Jongouk Choi, Hyunwoo Joe, and Changhee Jung. 2022. Capos: capacitor error resilience for energy harvesting systems. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 41, 11, 4539–4550.
- [17] Alexander Curtiss et al. 2021. Facebit: smart face masks platform. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 5, 4, 1–44.
- [18] Jasper De Winkel, Vito Kortbeek, Josiah Hester, and Przemyslaw Pawelczak. 2020. Battery-free game boy. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 4, 3, 1–34.
- [19] Alina M Deac et al. 2008. Bias-driven high-power microwave emission from mgo-based tunnel magnetoresistance devices. *Nature Physics*, 4, 10, 803–809.
- [20] Vishal Deep, Vishak Narayanan, Mathew Wymore, Daji Qiao, and Henry Duwe. 2020. HARC: A Heterogeneous Array of Redundant Persistent Clocks for Batteryless, Intermittently-Powered Systems. In *2020 IEEE Real-Time Systems Symposium (RTSS)*. 2020 IEEE Real-Time Systems Symposium (RTSS). (Dec. 2020), 270–282. doi:10.1109/RTSS49844.2020.00033.
- [21] Jan de Leeuw. 1977. Correctness of Kruskal's algorithms for monotone regression with ties. *Psychometrika*, 42, 1, (Mar. 1, 1977), 141–144. doi:10.1007/BF02293750.
- [22] Harsh Desai, Matteo Nardello, Davide Brunelli, and Brandon Lucia. 2022. Camaroptera: a long-range image sensor with local inference for remote sensing applications. *ACM Transactions on Embedded Computing Systems (TECS)*.
- [23] Jasper de Winkel, Carlo Delle Donne, Kasim Sinan Yildirim, Przemyslaw Pawelczak, and Josiah Hester. 2020. Reliable Timekeeping for Intermittent Computing. In *Proceedings of the Twenty-Fifth International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS '20)*. Association for Computing Machinery, New York, NY, USA, (Mar. 13, 2020), 53–67. ISBN: 978-1-4503-7102-5. doi:10.1145/3373376.3378464.
- [24] Maria Doglioni, Eren Yildiz, Matteo Nardello, Khakim Akhunov, Kasim Sinan Yildirim, and Davide Brunelli. 2025. Capdyn: adaptive self-scaling energy storage for powering batteryless iot. *ACM Transactions on Embedded Computing Systems*, 24, 5, 1–32.
- [25] Christian Duffee, Jordan Athas, Andrea Grimaldi, Deborah Volpe, Giovanni Finocchio, Ermin Wei, and Pedram Khalili Amiri. 2025. P-dits: probabilistic d-dimensional bits for extended-variable probabilistic computing. *Physical Review Applied*, 24, 4, 044077.
- [26] Christian Duffee, Jordan Athas, Igor Kadota, Randall Berry, and Pedram Khalili Amiri. 2026. P-dits for the frequency assignment problem with transmitter deactivation in dense environments. *IEEE Journal on Exploratory Solid-State Computational Devices and Circuits*.
- [27] Christian Duffee, Jordan Athas, Yixin Shao, Noraica Davila Melendez, Eleonora Raimondo, Jordan A Katine, Kerem Y Camsari, Giovanni Finocchio, and Pedram Khalili Amiri. 2025. An integrated-circuit-based probabilistic computer that uses voltage-controlled magnetic tunnel junctions as its entropy source. *Nature Electronics*, 8, 9, 784–793.
- [28] Christian Duffee, Chadbourne M Burling-Smith, Jordan Athas, Andrea Grimaldi, Giovanni Finocchio, Ermin Wei, and Pedram Khalili Amiri. 2026. P-dit probabilistic ising machine for solving the quadratic assignment problem. *arXiv preprint arXiv:2605.24408*.
- [29] Ferhat Erata, Eren Yildiz, Arda Goknil, Kasim Sinan Yildirim, Jakub Szefer, Ruzica Piskac, and Gokcin Sezgin. 2023. Etap: energy-aware timing analysis of intermittent programs. *ACM Transactions on Embedded Computing Systems*, 22, 2, 1–31.
- [30] Espressif Systems. 2025. ESP32-WROOM-32 Datasheet. Datasheet v3.6. Espressif Systems, (Aug. 8, 2025). https://documentation.espressif.com/esp32-wroom-32_datasheet_en.pdf.
- [31] F. N. Fritsch and J. Butland. 1984. A Method for Constructing Local Monotone Piecewise Cubic Interpolants. *SIAM Journal on Scientific and Statistical Computing*, 5, 2, (June 1984), 300–304. doi:10.1137/0905021.
- [32] Fernando Garcia-Redondo, Pranay Prabhat, and Mudit Bhargava. 2021. A Fokker-Planck Solver to Model MTJ Stochasticity. (June 23, 2021). arXiv: 2106.12304 [cs]. Retrieved Oct. 2, 2025 from <http://arxiv.org/abs/2106.12304>. Pre-published.
- [33] Fernando García-Redondo, Pranay Prabhat, Mudit Bhargava, and Cyrille Dray. 2021. A Compact Model for Scalable MTJ Simulation. (June 9, 2021). arXiv: 2106.04976 [cs]. Retrieved Oct. 2, 2025 from <http://arxiv.org/abs/2106.04976>. Pre-published.
- [34] Kai Geissdoerfer and Marco Zimmerling. 2021. Bootstrapping battery-free wireless networks: efficient neighbor discovery and synchronization in the face of intermittency. In *18th USENIX Symposium on Networked Systems Design and Implementation (NSDI 21)*, 439–455.
- [35] Kai Geissdoerfer and Marco Zimmerling. 2022. Learning to communicate effectively between battery-free devices. In *19th USENIX Symposium on Networked Systems Design and Implementation (NSDI 22)*, 419–435.
- [36] Kai Geissdoerfer and Marco Zimmerling. 2024. Riotee: an open-source hardware and software platform for the battery-free internet of things. In *Proceedings of the 22nd ACM Conference on Embedded Networked Sensor Systems*, 198–210.
- [37] Josiah Hester, Kevin Storer, and Jacob Sorber. 2017. Timely Execution on Intermittently Powered Batteryless Sensors. In *Proceedings of the 15th ACM Conference on Embedded Network Sensor Systems (SenSys '17)*. Association for Computing Machinery, New York, NY, USA, (Nov. 6, 2017), 1–13. ISBN: 978-1-4503-5459-2. doi:10.1145/3131672.3131673.
- [38] Josiah Hester, Kevin Storer, and Jacob Sorber. 2017. Timely execution on intermittently powered batteryless sensors. In *Proceedings of the 15th ACM Conference on Embedded Network Sensor Systems*, 1–13.
- [39] Josiah Hester, Nicole Tobias, Amir Rahmati, Lanny Sitanayah, Daniel Holcomb, Kevin Fu, Wayne P Burleson, and Jacob Sorber. 2016. Persistent clocks for batteryless sensing devices. *ACM Transactions on Embedded Computing Systems (TECS)*, 15, 4, 1–28.
- [40] Josiah Hester, Nicole Tobias, Amir Rahmati, Lanny Sitanayah, Daniel Holcomb, Kevin Fu, Wayne P. Burleson, and Jacob Sorber. 2016. Persistent Clocks for Batteryless Sensing Devices. *ACM Trans. Embed. Comput. Syst.*, 15, 4, (Aug. 2, 2016), 77:1–77:28. doi:10.1145/2903140.
- [41] Sumio Ikegawa, Frederick B Mancoff, and Sanjeev Aggarwal. 2021. Commercialization of mram—historical and future perspective. In *2021 IEEE International Interconnect Technology Conference (IITC)*. IEEE, 1–3.
- [42] Bashima Islam and Shahriar Nirjon. 2020. Scheduling computational and energy harvesting tasks in deadline-aware intermittent systems. In *2020 IEEE Real-Time and Embedded Technology and Applications Symposium (RTAS)*. IEEE, 95–109.
- [43] Bashima Islam and Shahriar Nirjon. 2019. Zygarde: time-sensitive on-device deep inference and adaptation on intermittently-powered systems. *arXiv preprint arXiv:1905.03854*.
- [44] Philo Juang, Hidekazu Oki, Yong Wang, Margaret Martonosi, Li-Shuan Peh, and Daniel Rubenstein. 2002. Energy-efficient computing for wildlife tracking: design tradeoffs and early experiences with ZebraNet. In *Proceedings of the 10th International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, 96–107.
- [45] Doo Hyung Kang and Mincheol Shin. 2021. Critical switching current density of magnetic tunnel junction with shape perpendicular magnetic anisotropy through the combination of spin-transfer and spin-orbit torques. *Scientific Reports*, 11, 1, (Nov. 24, 2021), 22842. doi:10.1038/s41598-021-02185-3.
- [46] Vito Kortbeek, Kasim Sinan Yildirim, Abu Bakar, Jacob Sorber, Josiah Hester, and Przemyslaw Pawelczak. 2020. Time-sensitive intermittent computing meets legacy software. In *Proceedings of the Twenty-Fifth International Conference*

- on Architectural Support for Programming Languages and Operating Systems, 85–99.
- [47] Jan de Leeuw, Kurt Hornik, and Patrick Mair. 2010. Isotone Optimization in R: Pool-Adjacent-Violators Algorithm (PAVA) and Active Set Methods. *Journal of Statistical Software*, 32, 1–24. doi:10.18637/jss.v032.i05.
- [48] Xiang Li et al. 2015. Thermally stable voltage-controlled perpendicular magnetic anisotropy in mo|cofeb| mgo structures. *Applied Physics Letters*, 107, 14.
- [49] Yunwen Liu and Jiang Xiao. 2025. Application of probabilistic bits in measurement and sensing. *Applied Physics Letters*, 126, 12, (Mar. 25, 2025), 122403. doi:10.1063/5.0253449.
- [50] Yunwen Liu and Jiang Xiao. 2025. Application of probabilistic bits in measurement and sensing. *Applied Physics Letters*, 126, 12.
- [51] LLVM Project. 2026. Llvm-mca - LLVM Machine Code Analyzer — LLVM 23.0.0git documentation. LLVM Compiler Infrastructure. (Apr. 14, 2026). Retrieved Apr. 14, 2026 from <https://llvm.org/docs/CommandGuide/llvm-mca.html>.
- [52] Kiwan Maeng, Alexei Colin, and Brandon Lucia. 2017. Alpaca: intermittent execution without checkpoints. *Proceedings of the ACM on Programming Languages*, 1, OOPSLA, 1–30.
- [53] Kiwan Maeng and Brandon Lucia. 2020. Adaptive low-overhead scheduling for periodic and reactive intermittent execution. In *Proceedings of the 41st ACM SIGPLAN Conference on Programming Language Design and Implementation*, 1005–1021.
- [54] Matthew J Marinella. 2021. Radiation effects in advanced and emerging non-volatile memories. *IEEE Transactions on Nuclear Science*, 68, 5, 546–572.
- [55] Andrew Markham, Niki Trigoni, Stephen A. Ellwood, and David W. Macdonald. 2010. Revealing the hidden lives of underground animals using magneto-inductive tracking. In *Proceedings of the 8th ACM Conference on Embedded Networked Sensor Systems (SenSys '10)*. Association for Computing Machinery, Zürich, Switzerland, 281–294. ISBN: 9781450303446. doi:10.1145/1869983.1870011.
- [56] Rohan Menon, Rohit Gujarathi, Ali Saffari, and Joshua R Smith. 2022. Wireless identification and sensing platform version 6.0. In *Proceedings of the 20th ACM Conference on Embedded Networked Sensor Systems*, 899–905.
- [57] Sébastien Mercier, Stéphane Villeneuve, Martin Mondor, and Ismail Uysal. 2017. Time–temperature management along the food cold chain: a review of recent developments. *Comprehensive Reviews in Food Science and Food Safety*, 16, 4, 647–667.
- [58] Cleve B. Moler. 2004. 3. Interpolation. In *Numerical Computing with Matlab*. Other Titles in Applied Mathematics. Society for Industrial and Applied Mathematics, (Jan. 2004), 93–116. ISBN: 978-0-89871-660-3. doi:10.1137/1.9780898717952.ch3.
- [59] Benjamin Ransford, Jacob Sorber, and Kevin Fu. 2011. Mementos: system support for long-running computation on rfid-scale devices. In *Proceedings of the sixteenth international conference on Architectural support for programming languages and operating systems*, 159–170.
- [60] Yixin Shao, Christian Duffee, Eleonora Raimondo, Noraica Davila, Victor Lopez-Dominguez, Jordan A Katine, Giovanni Finocchio, and Pedram Khalili Amiri. 2023. Probabilistic computing with voltage-controlled dynamics in magnetic tunnel junctions. *Nanotechnology*, 34, 49, 495203.
- [61] Nihal Sanjay Singh et al. 2024. Cmos plus stochastic nanomagnets enabling heterogeneous computers for probabilistic inference and learning. *Nature Communications*, 15, 1, 2685.
- [62] Robert Szewczyk, Alan Mainwaring, Joseph Polastre, John Anderson, and David Culler. 2004. An analysis of a large scale habitat monitoring application. In *Proceedings of the 2nd International Conference on Embedded Networked Sensor Systems (SenSys)*, 214–226.
- [63] Texas Instruments. 2018. MSP430FR596x, MSP430FR594x Mixed-Signal Microcontrollers. Datasheet SLAS704G. Texas Instruments, Dallas, Texas, U.S., (Aug. 2018). <https://www.ti.com/lit/ds/symlink/msp430fr5969.pdf>.
- [64] Evangelos Vasilakis. 2015. *An Instruction Level Energy Characterization of ARM Processors*. Technical Report. University of Crete, Foundation of Research and Technology Hellas, (Mar. 2015). Retrieved Apr. 14, 2026 from <https://projects.ic.s.forth.gr/carv/greenvm/files/tr450.pdf>.
- [65] KL Wang, JG Alzate, and P Khalili Amiri. 2013. Low-power non-volatile spintronic memory: stt-ram and beyond. *Journal of Physics D: Applied Physics*, 46, 7, 074003.
- [66] Harrison Williams and Matthew Hicks. 2024. Energy-adaptive buffering for efficient, responsive, and persistent batteryless systems. In *Proceedings of the 29th ACM International Conference on Architectural Support for Programming Languages and Operating Systems, Volume 3*, 268–282.
- [67] Eren Yildiz, Saad Ahmed, Bashima Islam, Josiah Hester, and Kasim Sinan Yildirim. 2023. Efficient and safe i/o operations for intermittent systems. In *Proceedings of the Eighteenth European Conference on Computer Systems*, 63–78.
- [68] Eren Yildiz, Khakim Akhunov, Lorenzo Antonio Riva, Arda Goknil, Ivan Kurtev, and Kasim Sinan Yildirim. 2024. Adaptable runtime monitoring for intermittent systems. In *Proceedings of the Nineteenth European Conference on Computer Systems*, 1175–1191.
- [69] Eren Yildiz, Davide Cavedon, Stefano Antonio Putelli, Josiah Hester, and Kasim Sinan Yildirim. 2026. Bionic: a co-designed hardware and runtime for time-sensitive battery-free iot. In *Proceedings of the 24th Annual International Conference on Mobile Systems, Applications and Services*, 324–337.
- [70] Eren Yildiz, Davide Cavedon, and Kasim Sinan Yildirim. 2024. On Tracking Time with Better Resolution and Range in Batteryless Systems. In *Proceedings of the 12th International Workshop on Energy Harvesting and Energy-Neutral Sensing Systems (ENSys '24)*. Association for Computing Machinery, New York, NY, USA, (Nov. 14, 2024), 42–47. ISBN: 979-8-4007-1296-8. doi:10.1145/3698384.3699617.
- [71] Kasim Sinan Yildirim, Amjad Yousef Majid, Dimitris Patoukas, Koen Schaper, Przemyslaw Pawelczak, and Josiah Hester. 2018. Ink: reactive kernel for tiny batteryless sensors. In *Proceedings of the 16th ACM Conference on Embedded Networked Sensor Systems*, 41–53.
- [72] Eren Yildiz, Lijun Chen, and Kasim Sinan Yildirim. 2022. Immortal threads: multithreaded event-driven intermittent computing on {ultra-low-power} microcontrollers. In *16th USENIX Symposium on Operating Systems Design and Implementation (OSDI 22)*, 339–355.
- [73] Eren Yildiz and Kasim Sinan Yildirim. 2021. Persistent timekeeping using harvested power measurements. In *Proceedings of the 19th ACM Conference on Embedded Networked Sensor Systems*, 572–574.
- [74] Wenzhe Zhang, Qiang Hao, and Gang Xiao. 2011. Low-frequency noise in serial arrays of mgo-based magnetic tunnel junctions. *Physical Review B—Condensed Matter and Materials Physics*, 84, 9, 094446.
- [75] Chen Zhao, Yen-Ting Wang, David Genzer, Degang Chen, and Randall Geiger. 2013. A CMOS on-chip temperature sensor with -0.21°C 0.17 °C inaccuracy from -20 °C to 100 °C. In *2013 IEEE International Symposium on Circuits and Systems (ISCAS)*. 2013 IEEE International Symposium on Circuits and Systems (ISCAS). (May 2013), 2621–2625. doi:10.1109/ISCAS.2013.6572416.

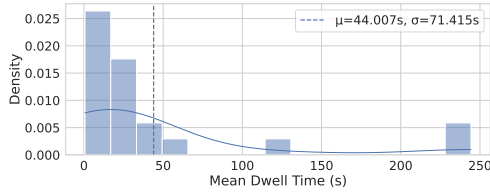


Figure A1: Mean dwell time distribution for 21 MTJ devices with a targeted 10 s dwell time.

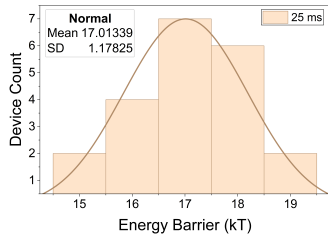


Figure A2: Energy barrier distribution for 21 MTJ devices with a targeted 25 ms dwell time.

A Simulator Validation

Because the design-space exploration in this work relies heavily on simulation, we verify that the simulator reproduces the behavior of experimentally measured devices. Fig. 10 compares the normalized array imbalance from the 21 fabricated devices against a Monte-Carlo simulation of a 64×256 FLINT array. Both datasets exhibit the same exponential decay, confirming that the simulator accurately captures the thermally activated Arrhenius switching dynamics $\tau \propto e^{E_b/k_b T}$ that govern FLINT operation. The curves begin to diverge after approximately 20 s because a finite ensemble of N devices approaches a non-zero statistical imbalance floor of $\sqrt{2/\pi N}$ (≈ 0.17 for $N=21$ versus ≈ 0.006 for the 16 384-device array). The plateau observed in experimental data therefore arises from finite-sample statistics rather than a systematic physical offset. This agreement between measurement and simulation provides confidence in the simulated design-space studies presented throughout this work.

B Further MTJ Quantification

Fig. A1 shows the distribution of mean dwell times for a targeted 10 s dwell time, reflecting the raw dwell times before conversion to energy barriers via the Néel-Arrhenius relation. Fig. A2 shows the energy barrier distribution for devices targeting a 25 ms dwell time. Together they demonstrate the tunability of the mean energy barrier: the normal distribution shifts to smaller energy barriers as the targeted dwell time decreases, and we expect a similarly predictable shift at other dwell-time targets.

C Further Single Array Simulation Results

ADC bit selection. In Fig. A3, we plot the timekeeper error with a varying number of ADC bits. We find that at least 10 bits are required to achieve timekeeping accuracy of at least 10%. For all

further experiments, we choose a 12-bit ADC because it is offered by many low-power microcontrollers [30, 63].

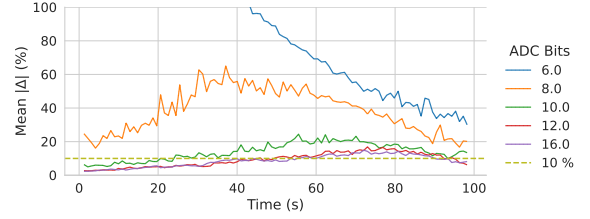


Figure A3: Timekeeper error with varying numbers of ADC bits.

Impact of Measurement Voltage. Finally, the voltage across the MTJs, among other intrinsic factors, determines the TMR ratio of the device. TMR drops when the voltage across the device increases, affecting the sensitivity of time estimation negatively; it is desirable to operate under low voltage conditions. We plot the timekeeper error as a function of increasing device voltages and appropriate prior calibration in Fig. A4. Up to a range of 0.5V per device, our calibration is able to achieve a 40-second range under 10% error. As noted prior, FLINT is designed to operate at 0.1 V across each device, keeping our operating point well under the upper limit and enabling larger TMR ratios and better time resolution.

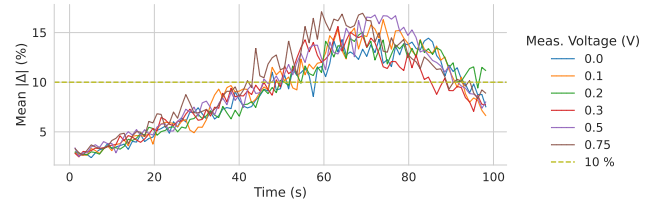


Figure A4: Timekeeper error with varying measurement voltages (with calibration).

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